PMSCS CSE660 Class Assignment: Software Effort Estimation

Course: Advanced Software Project Management **Topic:** Software Effort Estimation (Chapter 5)

Assignment Title: Balancing Act: Strategic Effort Estimation for a University-Wide Student Information System

Objective: The primary objective of this assignment is to assess your ability to critically analyze, apply, and synthesize different software estimation techniques in a realistic, multi-faceted project scenario. You will be required to move beyond simple calculation to justify your methodological choices, analyze risks, and propose a robust estimation strategy that accounts for the inherent uncertainties of a large-scale institutional software project.

Scenario: You are the lead project manager for a large, public university, "Jahangirnagar University." The university has decided to replace its legacy, monolithic Student Information System (SIS) with a modern, integrated, and web-based platform. The project is named "Project Polaris."

Project Polaris Overview: The new SIS must consolidate and streamline all student-related data and processes. It will consist of the following core modules:

1. **Student Portal (Web Application):**
    - A user-facing web interface for students to register for courses, view grades, pay fees, update personal information, and access academic records.
    - This module requires high usability, a responsive design, and integration with the university's authentication system (Single Sign-On).
2. **Administrative Backend (Business Logic & Workflow):**
    - A robust system for university staff (registrars, advisors, finance officers) to manage student records, course catalogs, enrollment rules, degree audits, financial aid, and generate official transcripts.
    - This module has complex business logic, workflow management, and requires strict role-based access control.
3. **Data Integration & Reporting Layer (Backend):**
    - A powerful data layer that handles the persistence of all SIS data. It must provide secure APIs for the portal and administrative modules.
    - It includes an enterprise data warehouse and a reporting engine to produce analytics for accreditation, state reporting, and institutional research, requiring complex data queries and aggregations.

- **Key Constraints:**
    - **Legacy Data Migration:** The system must successfully and accurately migrate over 20 years of student data from the old system to the new database.
    - **Time-to-Market:** The university's board has mandated a hard deadline. The new system must be fully operational and stable before the start of the Fall semester in **15 months**. A delay would cause enrollment chaos and severe reputational damage.

        - **Reliability & Security:** The SIS will handle sensitive Personally Identifiable Information (PII) and financial data. It must comply with Privacy Act of Bangladesh and other data privacy regulations, requiring stringent security and audit trails.
    - **Stakeholder Buy-in:** The project has a large and vocal group of stakeholders (faculty, department heads, student government, IT staff) with varying needs and priorities.
- **Available Resources:**
    - You have a core development team of 12 people. This team is a mix of:
        - Senior developers with deep experience in enterprise Java/.NET and web technologies.
        - Mid-level developers familiar with modern frameworks.
        - Junior developers and new graduates with limited professional experience but strong academic backgrounds.
    - The university's IT department has historical data on past internal projects (e.g., developing a new library management system, upgrading the HR portal). However, none of these projects approached the scale or complexity of Project Polaris.

Assignment Tasks:

Answer the following questions in a single, well-structured document (approx. 2000-2500 words, excluding references). Provide clear justifications for your arguments and reference specific concepts from the Chapter 5 presentation.

Part 1: Critical Analysis of Estimation Methods (40%)

Given the scenario of Project Polaris, critically evaluate the suitability of the following estimation methods. For each, discuss their specific strengths, weaknesses, and the potential risks if they were applied in isolation for this project.

1. **Bottom-Up Estimating:** The WBS-based approach is a classic method. What are the specific challenges of applying a purely bottom-up estimate to a project of this scale and with three distinct modules? Where might it be most useful?
2. **Top-Down (Parametric) Estimating:** Using a model like COCOMO requires a size input (e.g., KLOC) and productivity drivers. What are the critical challenges in applying COCOMO to the **Administrative Backend** module, given that much of its effort is in complex business logic rather than just "lines of code"?
3. **Analogy-Based Estimating:** This relies on historical data from past projects. Given the information provided, why would finding a single, valid source case for the entire Project Polaris be extremely difficult? What specific attributes of the project make it unique and unsuitable for a direct analogy?
4. **Parkinson's Law & Price-to-Win:** The 15-month deadline is a major constraint. Discuss the ethical and practical dangers of succumbing to a "Price-to-Win" strategy to secure

project approval or how "Parkinson's Law" might manifest if the estimate is inflated. How could these phenomena derail the project?

Part 2: Applying Parametric and Functional Size Models (40%)

For this part, you will apply the principles from Chapter 5 to generate initial estimates for specific components of Project Polaris and critically analyze their limitations.

2.1 COCOMO Application (for the Administrative Backend):

- You estimate the Administrative Backend to be approximately 85 KSLOC (Thousands of Source Lines of Code).
- Using the COCOMO constants provided in the slides for a **"semi-detached"** system (as it's a mix of new code and integration with old systems), calculate the **nominal effort** in staff-months. Note the constants: **Effort = 3.0 x (Size in KSLOC) ^ 1.12**.
- **Analysis:** Consider the following cost driver effects:
    - The development team's **programming language experience** is rated "low" (multiplier = 1.14).
    - The **product complexity** is rated "very high" (multiplier = 1.34).
    - The **required software reliability** is "high" (multiplier = 1.15).
    - Calculate the **adjusted effort** by multiplying the nominal estimate by all three multipliers. Show your calculation.
- **Interpretation:** What do these adjustments tell you about the major cost drivers for this module? Why is a purely algorithmic model like COCOMO likely insufficient on its own for this project?

2.2 Function Point Analysis (for the Student Portal):

- The Student Portal is well-suited for a functional size model. For one of its key transactions, "Register for a Course," the process involves the following:
    - The student inputs 8 data items (e.g., Student ID, Course Code, Section Number, Semester).
    - The system outputs 4 data items (e.g., Confirmation Number, Course Title, Schedule, Prerequisite Status).
    - The transaction accesses 7 different entity types (e.g., Student, Course, Section, Prerequisite, Schedule, Payment, Waitlist).
- Using the **Mark II Function Point** formula from the slides (**FP Count = (Ni * 0.58) + (Ne * 1.66) + (No * 0.26)**), calculate the size of this single transaction in Mark II FPs.
- **Discussion:** Compare the Mark II FP approach with the **COSMIC FP** approach described in the chapter. Which method (Mark II or COSMIC) would be more appropriate for the **Data Integration & Reporting Layer**? Justify your choice, specifically mentioning the types of data movements involved.

Part 3: Synthesis and Recommendation (20%)

Based on your analysis in Parts 1 and 2, propose a comprehensive, hybrid estimation strategy for Project Polaris.

1. **Recommend a Hybrid Approach:**
    - Propose a strategy that combines at least two of the methods discussed (e.g., Top-Down for overall budget, Bottom-Up for high-risk data migration, Analogy for the reporting engine).
    - Explain in detail *why* you are choosing specific methods for specific modules. For example, why might you use a bottom-up estimate for the legacy data migration?
2. **Address Specific Risks:**
    - Explain how your recommended strategy directly helps mitigate the unique risks of Project Polaris, specifically:
        - The hard 15-month deadline.
        - The complexity and risk associated with legacy data migration.
        - The lack of directly comparable historical projects for the entire system.
        - The high reliability and FERPA compliance requirements.
3. **Reviewing the Estimate:**
    - Finally, imagine you have produced a draft estimate (e.g., a total of 150 staff-months). Using the "how to review estimates" questions from the concluding slide of Chapter 5, provide a detailed critique of this hypothetical estimate.
    - Answer each of the four questions from the slide, using specific examples and details from the Project Polaris scenario to illustrate your points.
